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“Stop Building AI Agents — Start Building AI Skills”

Ernest Bonat, Ph.D.
Senior AI Engineer

1. “Agents should orchestrate. Skills should perform the work”

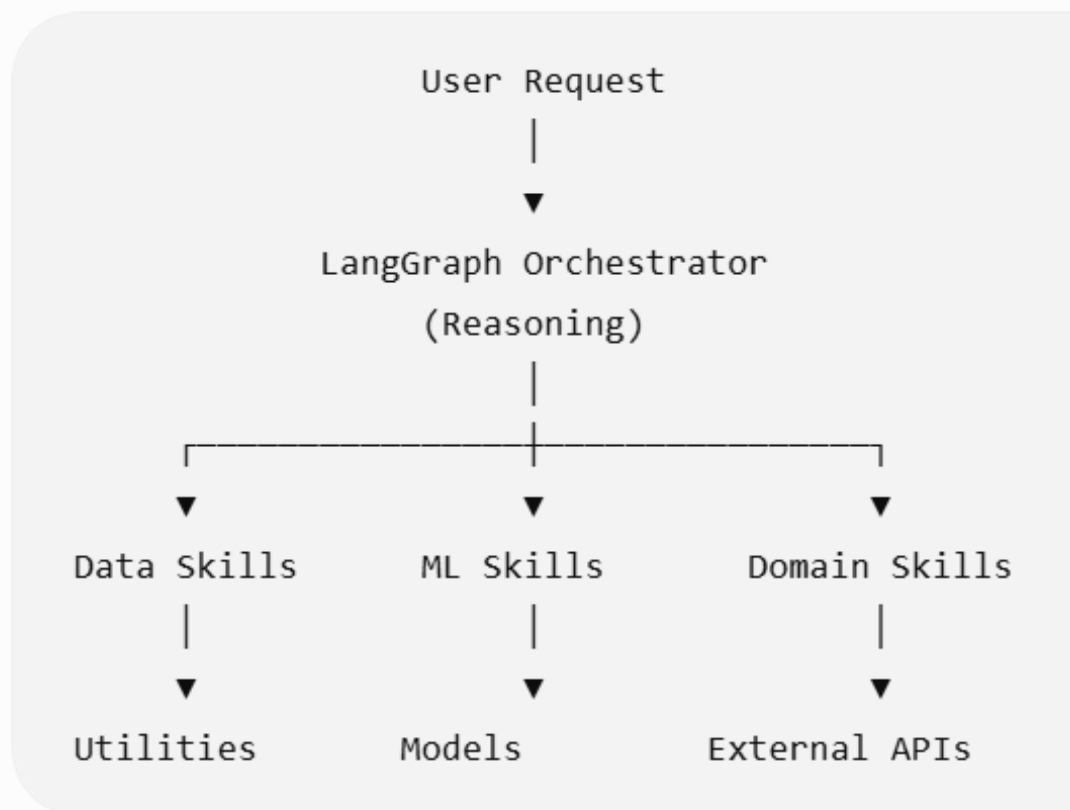
Most organizations today - build **monolithic AI agents** that combine reasoning, business logic, database access, API calls, and Machine Learning into a single implementation.

The skills (SKILLS) as first-class software component.

An Agent becomes a **lightweight orchestrator** responsible for planning, selecting, and coordinating reusable skills.

"Agent = Orchestrator; Skill = Capability"

2. Architecture



The orchestrator is intentionally **kept small**. Everything else becomes reusable.

3. What is a Skill?

A Skill is a reusable software component that performs **one well-defined task**.

Examples: Data preprocessing, Feature engineering, SQL querying, Report generation, etc.

A skill should never contain planning logic.

4. Agent Responsibilities

- Understand the user's objective.
- Plan the workflow.
- Select the appropriate skills.
- Pass state between skills.
- Validate outputs.
- Produce the final response.

5. Skill Responsibilities

- Perform one task extremely well.
- Receive typed input.
- Produce typed output.
- Be deterministic whenever possible.
- Be independently testable.
- **Be reusable by many agents.**

6. Project Structure

```
project/
```

```
README.md
```

```
CLAUDE.md
```

```
pyproject.toml
```

```
requirements.txt
```

```
knowledge/
```

```
    architecture.md
```

```
    coding_standards.md
```

```
    prompts.md
```

```
agents/
```

```
    orchestrator.py
```

```
skills/
```

```
    preprocessing/
```

```
        skill.py
```

```
        README.md
```

```
        tests/
```

```
    feature_engineering/
```

```
        skill.py
```


model_training/

skill.py

evaluation/

skill.py

reporting/

skill.py

visualization/

skill.py

pubmed/

skill.py

rag/

skill.py

models/

tests/

docs/

7. LangGraph Architecture



Every node is either: Planner, Router, Validator and Skill

8. SKILLS Benefits

- **Reusability:** One preprocessing skill. Used by 20 agents.
- **Maintainability:** Bug fix once. All agents improve.
- **Testing:** Each skill independently testable.
- **Explainability:** Every execution step is visible.
- **Scalability:** Create Skill -> Register Skill -> All agents can use it.

9. Conclusion

- Skills are your business logic, and agents are your orchestration logic.
- Keeping those two concerns separate makes systems easier to maintain, test, and evolve.

10. Simple Demo

Create a reusable skill that calculates a person's **Body Mass Index** (BMI). Then demonstrate how this skill can be reused by multiple AI agents, such as a Health Assessment Agent, a Nutrition Advisor Agent, and a Fitness Coach Agent.